

ARTICLE 16

MAGNETIC FLUX LEAKAGE (MFL) EXAMINATION

T-1610 SCOPE

This Article describes the Magnetic Flux Leakage (MFL) examination method requirements applicable for performing MFL examinations on coated and uncoated ferromagnetic materials from one surface. MFL is used in the examination of tube and piping to find unwelded areas of longitudinal weld joints. It is also used as a post construction examination method to evaluate the condition of plate materials, such as storage tank floors, and piping for corrosion or other forms of degradation. Other imperfections that may be detected are cracks, seams, incomplete fusion, incomplete penetration, dents, laps, and nonmetallic inclusions, etc.

When this Article is specified by a referencing Code Section, the MFL method described in this Article shall be used together with Article 1, General Requirements.

T-1620 GENERAL

T-1621 Personnel Qualification Requirements

The user of this Article shall be responsible for documented training, qualification, and certification of personnel performing MFL examination. Personnel performing supplemental examinations, such as ultrasonic (UT) examinations, shall be qualified in accordance with the referencing Code Section.

T-1622 Equipment Qualification Requirements

The equipment operation shall be demonstrated by successfully completing the unit verification and function tests outlined as follows.

T-1622.1 Reference Specimen. All MFL examinations shall have a reference plate or pipe section to ensure the equipment is performing in accordance with the manufacturer's specifications prior to use. The reference specimen for plate shall consist of a plate that is made from a material of the same nominal thickness, product form, and composition as the component to be examined. The plate specimen shall have notches or other discontinuities machined into the bottom of the plate, as shown in Fig. T-1622.1.1. The reference specimen for pipe or tubing shall consist of a

pipe or tube that is made from a material of the same nominal pipe or tube sizes, product form, and composition as the component to be examined. The pipe or tube specimen shall have notch discontinuities machined into the inside and outside surfaces as shown in Fig. T-1622.1.2. The depths and widths of the artificial discontinuities should be similar to the sizes and physical characteristics of discontinuities to be detected. If nonmagnetic coatings or temporary coverings will be present during the examination, the reference specimen shall be coated or covered with the nonmagnetic coatings or covers representative of the maximum thickness that will be encountered during the examination.

T-1622.2 System Verification and Function Checks.

The manufacturer's verification procedure shall be conducted initially to ensure that the system is functioning as designed. The functional check shall be made by scanning the reference plate over the range of scanning speeds to be utilized during the examination. Equipment settings shall be documented.

T-1622.3 Performance Confirmation. A functional check shall be conducted at the beginning and end of each examination, every 8 hr, or when equipment has malfunctioned and been repaired. If it is determined that the equipment is not functioning properly, needed adjustments shall be made and all areas examined since the last performance check shall be reexamined.

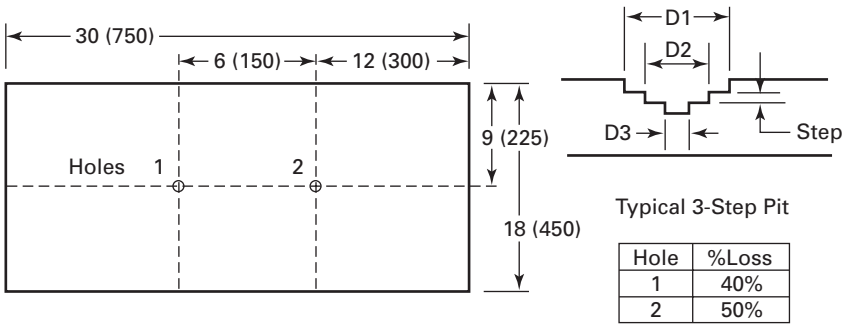
T-1623 Written Procedure Requirements

T-1623.1 Requirements. MFL examination shall be performed in accordance with a written procedure that shall, as a minimum, contain the requirements listed in Table T-1623. The written procedure shall establish a single value, or range of values, for each requirement.

The procedure shall address, as a minimum, the identification of imperfections, reference materials used to set up equipment, location and mapping of imperfections, and the extent of coverage. The procedure shall address the field strength of the magnets, the functioning of the sensors, and the operation of the signal-processing unit. Other examination methods that will be used to supplement the MFL examination shall be identified in the procedure.

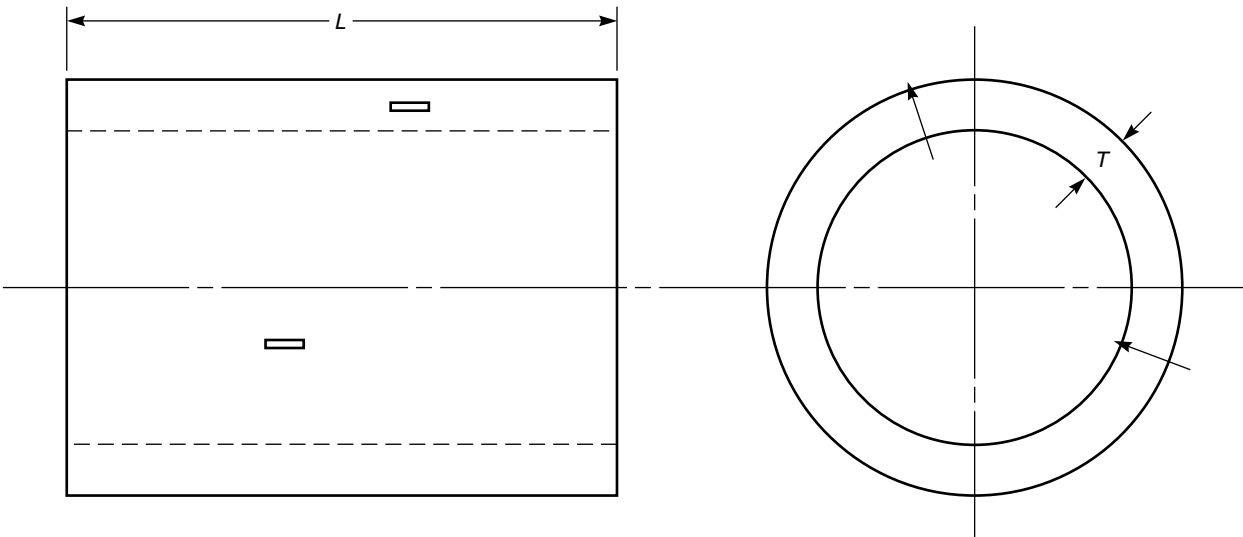
FIG. T-1622.1.1 REFERENCE PLATE DIMENSIONS

Plate Thickness	Hole Number	Number of Steps	Step Size	Diameter D1	Diameter D2	Diameter D3	Diameter D4	Diameter D5
$\frac{1}{4}$ (6)	1	3	.032 (0.8)	.47 (12)	.32 (8)	.12 (3)		
	2	4	.032 (0.8)	.62 (16)	.47 (12)	.32 (8)	.12 (3)	
$\frac{5}{16}$ (8)	1	4	.032 (0.8)	.62 (16)	.47 (12)	.32 (8)	.16 (4)	
	2	5	.032 (0.8)	.78 (20)	.62 (16)	.47 (12)	.32 (8)	.16 (4)
$\frac{3}{8}$ (10)	1	4	.039 (1)	.78 (20)	.59 (15)	.39 (10)	.2 (5)	
	2	5	.039 (1)	.96 (24)	.78 (20)	.59 (15)	.39 (10)	.2 (5)



GENERAL NOTE: Dimensions of references are in in. (mm).

FIG. T-1622.1.2 REFERENCE PIPE OR TUBE DIMENSIONS



Typical Block Dimensions

Minimum length L 8 in. (200 mm) or $8T$,
whichever is greater
Full circumference

Specific Notch Dimensions

Length L – 1 in. (25 mm) maximum
Depth D – 10% T with tolerance
(+10% – 20%) of depth
Width – 0.010 in. (0.25 mm) maximum
Location – not closer than $3T$ from any
block edge or other notch in axial direction
Minimum 90 deg from adjacent notch(es)

TABLE T-1623
REQUIREMENTS OF AN MFL EXAMINATION
PROCEDURE

Requirement	Essential Variable	Non-Essential Variable
Equipment manufacturer/model	X	...
Sensor type: manufacturer and model	X	...
Scanning speed/speed range	X	...
Overlap	X	...
Lift-off	X	...
Material examined	X	...
Material thickness range and dimensions	X	...
Reference specimen and calibration materials	X	...
Software	X	...
Evaluation of indications	X	...
Surface conditioning	X	...
Coating/sheet thickness	X	...
Performance demonstration requirements, when required	X	...
Scanning technique (remote control/manual)	...	X
Scanning equipment/fixtures	...	X
Personnel qualification requirements	...	X

T-1623.2 Procedure Qualification. When procedure qualification is specified, a change of a requirement in Table T-1623 identified as an essential variable shall require requalification of the written procedure by demonstration. A change in a requirement identified as a nonessential variable does not require requalification of the written procedure. All changes of essential or nonessential variables from those specified within the written procedure shall require revision of, or an addendum to, the written procedure.

T-1630 EQUIPMENT

The equipment shall consist of magnets, sensor or sensor array, and related electronic circuitry. A reference indicator, such as a ruled scale or linear array of illuminated light-emitting diodes, should be used to provide a means for identifying the approximate lateral position of indications. The equipment may be designed for manual scanning or may be motor driven. Software may be incorporated to assist in detection and characterization of discontinuities.

T-1640 REQUIREMENTS

(a) The surface shall be cleaned of all loose scale and debris that could interfere with the examination and movement of the scanner. The surface should be sufficiently flat to minimize excessive changes in lift-off and vibration.

Alternate techniques will be required to handle variables exceeding those specified in the procedure.

(b) Cleaning may be accomplished using high-pressure water blast or by sandblasting. If the material is coated and the coating is not removed, it shall be demonstrated that the MFL equipment can detect the specified imperfections through the maximum thickness of the temporary sheet or coating.

(c) If a temporary sheet or coating is applied between the scanner and plate to provide a smooth surface, for example, on a heavily pitted surface, it shall be demonstrated that the equipment can find the specified imperfections through the maximum thickness of the temporary sheet or coating.

T-1650 CALIBRATION

The MFL equipment shall be recalibrated annually and whenever the equipment is subjected to major damage following required repairs. If equipment has not been in use for 1 year or more, calibration shall be done prior to first use.

T-1660 EXAMINATION

(a) Areas to be examined shall be scanned in accordance with a written procedure. Each pass of the sensing unit shall be overlapped in accordance with the written procedure.

(b) The unit shall be scanned manually or by a motor-driven system. Other examination methods may be used to provide coverage in areas not accessible to MFL examinations, in accordance with the written procedure. Typical examples of inaccessible areas in storage tanks are lap welds and corner welds adjacent to the shell or other obstructions, such as roof columns and sumps.

(c) Imperfections detected with MFL exceeding the acceptance standard signal shall be confirmed by supplemental examination(s) or be rejected. Supplemental examination shall be performed in accordance with written procedures.

(d) Where detection of linear imperfections is required, an additional scan shall be performed in a direction approximately perpendicular to the initial scanning direction.

T-1670 EVALUATION

All indications shall be evaluated in accordance with the referencing Code Section.

T-1680 DOCUMENTATION

A report of the examination shall contain the following information:

- (a) plate material specification, nominal wall thickness, pipe diameter, as applicable;
- (b) description, such as drawing/sketches, documenting areas examined, and/or areas inaccessible;
- (c) identification of the procedure used for the examination;
- (d) system detection sensitivity (minimum size of imperfections detectable);
- (e) location, depth, and type of all imperfections that meet or exceed the reporting criteria;

- (f) examination personnel identity and, when required by referencing Code Section, qualification level;
- (g) model and serial number of equipment utilized for the examination, including supplemental equipment;
- (h) date and time of examination;
- (i) date and time of performance verification checks; and
- (j) supplemental methods utilized and reference to associated reports.